Experiment: File Type Detection using Magic Numbers

Aim

To identify the true type of a file by reading its magic number (file signature) using Python.

Algorithm

1. Create a dictionary containing common file signatures (magic numbers) and their corresponding file types.
2. Read the file path from the user.
3. Open the file in binary mode and read the first 8 bytes (header).
4. Compare the file header with the stored magic number signatures.
5. Display the detected file type if a match is found; otherwise, display "Unknown File Type".

Python Code

```
# magic_number_detector.py


magic_numbers = {

    b'\x89PNG\r\n\x1a\n': "PNG Image",

    b'\xFF\xD8\xFF': "JPEG Image",

    b'%PDF': "PDF Document",

    b'PK\x03\x04': "ZIP/DOCX/XLSX/PPTX File",

    b'GIF87a': "GIF Image",

    b'GIF89a': "GIF Image"

}
```

```
filename = input("Enter file path: ").strip().strip('"')

try:
    with open(filename, "rb") as file:
        header = file.read(8)

    found = False

    for signature, filetype in magic_numbers.items():
        if header.startswith(signature):
            print("\nFile Type:", filetype)
            found = True
            break

    if not found:
        print("\nUnknown File Type")

except FileNotFoundError:
    print("File not found!")
except Exception as e:
    print("Error:", e)
```

Sample Input

Enter file path:

C:\Users\saich\OneDrive\Desktop\Sai Charan Tej_Resume.pdf

Sample Output

File Type: PDF Document

Result

The program successfully detected the actual file type by reading the file's magic number (header). It identified the selected file as a PDF Document, demonstrating that file signatures can accurately determine a file's true format regardless of its file extension.